

Attributing Mental States to Non-Embodied Autonomous Systems: A Systematic Review

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Abstract

Rapidly evolving, non-embodied autonomous agents, oftentimes powered by AI, have been shown to trigger mental state attributions by users. However, unlike embodied agents, attributions to non-embodied autonomous agents are not caused by physical presence but by language or abstract interfaces. To better understand this fast-growing field, the present article presents a systematic literature review of mental state attribution to non-embodied autonomous agents. Based on a literature body of 400+ papers, 26 relevant publications were identified, providing an overview of the communities studying mental state attributions, the study's design choices, and findings of these efforts. The findings indicate that (1) the HCI community is less involved in the study of mental state attributions, (2) a lacking validity of the measures limits the current measurement of mental state attributions, (3) there is a skew towards chat- and voice bots, and (4) mental state attributions increased vulnerabilities of users and shifts in accountability.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in HCI**; • **General and reference** → **Surveys and overviews**; **General literature**.

Keywords

mental states, mind perception, non-embodied, autonomous, agent, artificial intelligence

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1 Introduction & Related Work

Mental state attribution refers to the ability to ascribe thoughts, feelings, beliefs, intentions, and other mental states to oneself and others, oftentimes with the aim to understand and predict their behavior [14, 35]. As such, mental state attribution is based on the

assumptions that the other can act (has agency) and has experiences (thoughts and feelings) [10, 14]. This process is central to human social interactions, enabling people to navigate complex social environments and interact with others effectively. Consequently, attributing mental states to an agent influences how individuals perceive and interact with the world.

Previous research could show that humans do not only attribute mental states to other humans but also animals (see, e.g., [8, 14]), and inanimate things such as geometrical shapes [1]. Likewise, it was also found that humans commonly attribute mental states to technologies, predominantly embodied agents such as (social) robots (see [42] for a review). Results of these efforts show that understanding mental state attribution is important because it has significant implications for how humans engage with technology. For example, it was found that when users attribute mental states to robots, they are more likely to trust [9], show moral concern for [15], collaborate with [44], and feel empathy toward these agents [23].

Given that physical presence (i.e., embodiment) favors mental state attributions [27] to agents, it is no surprise that a majority of studies focuses on mental state attributions to embodied agents such as (social) robots. However, with the rise of (generative) AI, there has been a shift toward examining mental state attribution in agents that lack physical presence, but instead interact with users through language or abstract interfaces. As (generative) AI becomes increasingly pervasive in domains of everyday life, understanding how and with which consequences people attribute mental states to these non-embodied, autonomous agents is becoming an increasingly critical area of study. Moreover, as the field of non-embodied, autonomous agents is rapidly growing, it is important to examine which academic disciplines engage with the topic of mental state attribution, to better identify both the breadth of inquiry and the specific approaches taken across different fields. This mapping process can reveal potential gaps in the literature or areas where cross-disciplinary collaboration may yield novel insights. Taken together, this systematic review answers the following three research questions:

- RQ1: Which scientific communities investigate the attribution of mental states to non-embodied, autonomous agents?
- RQ2: What methods have been used in studies of mental state attribution to non-embodied, autonomous agents in terms of samples, stimulus materials, and measures?
- RQ3: What are the previous findings on mental state attribution to non-embodied, autonomous agents in terms of causes, consequences, and differential findings between different agents?

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To answer these needs, by summarizing existing empirical evidence, this literature review provides an extensive overview of the state-of-the-art of mental state attributions to non-embodied, autonomous agents. In doing so, this work provides both, an overview of communities that have examined mental state attributions, as well as a systematic summary of study methods (sample, type of study, stimulus material, measurements), and empirical findings (causes & consequence of mental state attributions).

2 Methods

To ensure a high quality, this systematic review follows the Statement on Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020 Statement) [32]. A systematic overview of all steps pursued can be found in the online supplementary material.

2.1 Scope and inclusion criteria

For this review, the following inclusion criteria were developed:

(1) Peer Review and Publication: The paper must have been peer-reviewed and published in an academic journal or conference proceedings. Preprints and theses were excluded. **(2) Language:** The paper had to be published in English. **Empirical Research with Human Subjects:** The paper must have been empirical (experimental or observational) and involve human subjects. Studies based solely on simulation data and qualitative studies were excluded. Papers that introduced a theory, a framework, or guidelines without empirical data, reviews, and meta-analyses, were also excluded. **(3) Mental State Attribution to an autonomous agent:** The paper had to include an evaluation or analysis of mental state attribution in the context of non-embodied, autonomous agents.

The following keywords were used to search for papers including the central constructs. Keywords had to be present in either the title, the abstract, or as a keyword: **Autonomous Agents:** "artificial intelligence" OR "machine learning" OR "deep learning" OR algorithm OR automation OR autonomous OR "conversational agent" OR computer OR machine OR technology OR bot. As **mental states** can be referred to using various terms, this keyword search was extended by: AND "intentional stance" OR "mind perception" OR "theory of mind" OR "machine consciousness" OR "mental state attribution". To cover the **attribution** of mental states AND attri* OR ascri* OR perce* were included. Because this review only focuses on non-embodied AI, the search was completed by AND NOT robot*.

2.2 Search Strategies and Selection Process

To select suitable papers, the following platforms were queried on August, 4 & 5, 2024: the ACM library, and Scopus. The search was finalized by examining and cross-referencing relevant citations from the papers found. Hence, the time frame of the search ranged from the date of database inception to August 5, 2024.

The database search resulted in 482 papers. A single coder removed $n = 41$ duplicates, extended abstracts, and work that was not peer-reviewed (preprints & theses). In addition, nine papers were added after cross-referencing. After this initial selection, two coders reviewed all remaining papers' titles and abstracts for the inclusion criteria. After this step, 85 papers remained that were

read in-depth by both coders to ascertain the inclusion of the manuscript. The coders worked independently and reached a satisfactory inter-rater agreement of Krippendorff's $\alpha = .83$ for inclusion of articles after a first round of coding. Disagreements were resolved via consensus discussions. From the over 400 papers found, 26 papers were included in this survey. A graphical overview of the selection process, can be found in Figure 1 in the Appendix. Studies were published during the years 2017 and 2024.

All articles selected for final inclusion were copied into a spreadsheet and coded by a single coder in reference to the primary categories of interest: publication type (journal or proceedings), outlet, outlet disciplines, agent type, task, experimental manipulation, assessment of mental state attribution, finding type (cause, consequence, or differential), and finding. In addition to these categories, data to describe the datasets, such as the study participants' background, the number of participants, and the general set-up of the study (online, lab, or field) was also collected. The spreadsheet containing the list of all selected papers and their respective coding is publicly accessible in the online supplementary material.

3 Results

3.1 RQ1: Which scientific communities investigate the attribution of mental states to non-embodied, autonomous agents?

To get a better understanding of the communities that study mental state attribution, data on the outlets that published the papers included in this review was collected. As a first step, the publications were divided into two categories: (1) journal publication & book chapters, and (2) conference proceedings. It was found that the majority ($n = 20$) of papers were published in scientific journals or as a book chapter. Among the remaining papers, four were published in conference proceedings, and one was published at a workshop.

Having a closer look at the outlets, all publications were sorted into disciplinary categories, using Scimago's Journal & Country Rank¹. Results show that eleven of the reviewed papers were published in HCI-related outlets such as *Computers in Human Behavior* or the *International Journal of Human-Computer Interaction*. Notably, all four articles in conference proceedings were among these eleven publications.

A second cluster of disciplines ($n = 8$) that emerged was journals that published work related to marketing, management, and information systems such as the *International Journal of Information Management*, *Psychology and Marketing*, and *Journal of Marketing*. Slightly overlapping with these two clusters, a third cluster comprised articles from psychology and communication outlets ($n = 11$) such as *Communication Research*, *New Media & Society*, and the *Journal of Experimental Psychology: General*.

In sum, it seems that, while the issue of mental state attribution has been picked up to some extent by the HCI community, the topic is predominantly discussed in other communities such as marketing and consumer research, communications, and psychology. Particularly, CHI contributed only one publication to the body of reviewed articles (see [38]).

¹<https://www.scimagojr.com/>

3.2 RQ2: What methods have been used in studies of mental state attribution to non-embodied, autonomous agents in terms of samples, stimulus materials, and measures?

The Samples: Of the 26 articles included in this review, the majority (24) were online studies, while the six were conducted in the lab, and one paper reported an additional field study (see [36] (higher study count than papers included in the review is a result of some articles reporting multiple studies). The sample size of the studies ranged from 12 (see [34] to 640 (see [36]) participants. Where information was available, the studies disclosed to have recruited participants from various crowd-sourcing platforms such as Amazon Mechanical Turk or Prolific (14 in total) or local equivalents (e.g., Sojump, Wenjuanxing, & Credamo for Chinese samples). Otherwise, six studies reported student samples and two convenience samples.

The agents: The autonomous agents in the reviewed studies covered a wide spectrum of applications and contexts. However, a majority of studies examined voice- ($n = 9$) or chat-bots ($n = 8$). As such, voice bots were predominantly studied in their function as digital personal assistants/virtual partners [4, 5, 21, 22, 29, 34, 36, 38], or included as a smart home device [17]. Two studies explicitly investigated perceptions of Alexa [34, 38]. Concerning chat-bots, in most studies, chat-bots were simply introduced as chat-bots [18, 24, 26, 28, 30, 36, 38]. Yet, one study explicitly examined chat-bots as social companions [16], and two studies investigated mental state attribution to ChatGPT [7, 38]. In one instance, a study examined the experience with voice- and chat-bots, summarizing both under the umbrella term conversational AI [22].

Besides these rather explicit cases, the agents were less defined, discussing, for example, artificial autonomous actors (A-bots) [2], technology [37], or a machine powered by AI [25]. In one instance, this choice was very deliberate as the authors of the study were interested in mental state attributions to autonomous systems in general [11]. Three studies employed terms such as algorithm [41] or AI [19, 39] to describe the agent. However, in both studies, the authors further defined the use cases of the algorithm and AI, respectively, setting contexts such as financial regulation, advertisement, driving, social media rankings, student admission, and product personalization (see [41]), or farming management [39] and parole sentencing, content moderation, social media bot, and advertisement [41]. Notably, besides the two instances of ChatGPT for the chat-bot category, only one other study examined mental state attribution to generative AI [31].

The tasks: Many studies included in this review used vignette settings to examine mental state attributions [2, 29, 38, 39, 41], meaning that participants did not interact with the respective agent but either read about it or saw a video of the agent. While limiting ecological validity, vignette studies pose significant advantages, such as isolating variables. Moreover, various studies either asked participants about their past experiences with certain agents [5, 7, 16, 20, 21, 34], or about imagined experiences [17], allowing only for correlational instead of causal evidence (see more in the findings section).

In contrast to this, several studies let participants interact with agents directly. For example, Lee et al. [26] let participants converse with a chat-bot about purchasing sneakers, or Liao et al. [30] who let participants either give or receive advice from a chat-bot. At least two studies made use of economic games, letting participants interact with agents in the matching pennies games [4] or the ultimatum and dictator game [25]. In a similar vein, Heyselaar and Bosse [18] employed a referential communication game to assess users' mental state attribution. Schroeder and Epley [36] asked participants to listen to a text and determine whether it was created by a human or a machine, while Schweitzer and Waytz [37] asked participants to describe how machines and humans differ.

Assessment of mental state attribution: To assess users' mental state attributions, most studies employed self-report measures in the form of Likert-Scales. To that end, following theoretical considerations by Gray et al. [14] and Fiske et al. [10], various studies operationalized mental state attribution either along the dimensions of perceived agency and perceived experience or perceived warmth and perceived competence. In doing so, studies used items either following Gray et al. [14] and Fiske et al. [10] directly, or referenced works that previously employed Gray et al. [14] and Fiske et al. [10], such as Wang and Krumhuber [43] and Lee et al. [26].

In addition to these theoretically grounded operationalizations of mental state attribution, additional constructs were also employed. Using self-developed items Ashton and Franklin [2] assessed users' perceptions of the agent's capabilities to experience pain and passion, the agent's ascribed legal rights, and parenting qualities, whereas Colombatto and Fleming [7], Scott et al. [38] developed items to assess user's perceptions of the capability of an agent's conscious experience. Going in a similar direction, Hindennach et al. [19] asked users to rate the perceived awareness of the AI system, but also assessed the perceived wrongness and responsibility of the agent.

Relying on existing scales, Li et al. [28] measured the perceived functional communicative responsibility by Aune et al. [3], Goldberg et al. [13] and the perceived emotional communicative responsibility using Aune et al. [3], Gelbrich et al. [12]. Likewise, Li et al. [29] used a scale to measure the agent's perceived ability to emphasize using Shen [40].

Importantly, neither measures grounded in theory, self-developed items, nor items/scales from other studies were validated. This is concerning as these items and scales lack established psychometric properties, leading to potential biases, reduced generalizability, and difficulties in replicating findings, ultimately undermining the scientific rigor and credibility of the research conducted. Moreover, it is also noteworthy that there seemed to be a heavy skew towards self-report measures. Only four works were identified that did not use some sort of Likert-Scale type questions, but instead employed an implicit measure developed by Banks [6] or dictionary approaches and natural language processing [17, 18, 37].

Particularly, the focus of two studies was to introduce new measures of mental state attribution. As such, Hartmann et al. [17] developed the language model MindMiner. Combining deep learning with interpretable text-mining methods, the authors offer an approach to assess mental state attribution from unstructured text. Likewise, Schweitzer and Waytz [37] developed the text analysis

tool—the Mind Perception Dictionary—which is based on the Linguistic Inquiry Word Count software [33].

Often, measures of mental state attributions were accompanied by related constructs. For an overview of these additional measures, see Appendix 6.2.

3.3 RQ3: What are the previous findings on mental state attribution to non-embodied, autonomous agents?

The following three subsections provide a rather general overview of common trends and possible conflicting results, instead of in-depth insights into all studies' practical and theoretical implications as well as their limitations².

Causes of mental state attribution: Overall, at least 8 papers reported on the causes of mental state attributions to non-embodied autonomous agents. To that end, different subclasses were identified: (1) characteristics of the agent, (2) contextual cues, and (3) individual differences. *Characteristics of the agent:* At least three papers indicated that the language employed by an agent affected users' mental state attributions. To that end, Hu and Pan [22] found that conversational agents that employed certain linguistic strategies were more forgivable. The authors found that, compared to a more literal language style, cute language styles induced more mental state attributions. Likewise, Schroeder and Epley [36] found that adding a voice to computer-generated text increased the tendency to anthropomorphize the agent. However, the authors also found that adding a voice alone was not enough. The most important cue to induce mental state attributions was the intonation of the voice. Similarly, paralinguistic cues positively impacted mental state attributions, which, in turn, increased perceptions of both co-presence and closeness [26].

Going beyond, linguistic cues, Li et al. [28] found that perceived intelligence and perceived reliability both had a positive effect on the perceived functional communicative responsibility of an agent. Additionally, the authors found that perceived intimacy and perceived dependence increased the perceived emotional communicative responsibility. Affecting mental state attribution depending on the dimension of mental state attribution, Frischknecht [11] found that, while increased levels of technical autonomy increased the perception of an agent's competence and agency, it did not affect perceptions of warmth and experience.

Contextual cues: Only one study examined how contextual cues affected mental state attributions. Hartmann et al. [17] found that envisioning a more communal (vs. instrumental) relationship with a consumer-smart object increased their participants' mental state attributions to the object.

Individual differences: The following studies examined how variables related to the users affected mental state attributions. One study revealed that people who are more likely to anthropomorphise were also more inclined to perceive a mind in an AI teleseller [29]. With mental state attribution being a sub-field of anthropomorphism, this result is less surprising. More surprisingly, Shank and DeSanti [39] found, for the context of moral violations, that

knowledge of the AI's algorithm increased reporting of the AI to have a mind. The authors explain this finding by referring to the subdimension of mental state attribution, perceived agency. They speculate that knowing the algorithm might have increased perceptions of its capacity to plan and act with intentions.

Consequences of mind perception: Eight studies reported the consequences of attributing mental states to a non-embodied autonomous agent. To that end, various studies could connect mental state attribution to affect how users related to the agents. Examining the acceptance of social advice provided by chatbots, Lee and Hahn [24] found that implicit and explicit mental state attribution to a chatbot increased the perceived message effectiveness of the social advice. Complementing this, only when their participants did not implicitly attribute mental states to the chatbot, providing emotional support decreased the extent to which the participants considered the chatbots' messages as effective [24]. Likewise, Lee et al. [26] found that the more participants attributed mental states to a chatbot, the stronger they sensed co-presence and closeness, and the more they wanted to use the chatbot in the future. Investigating economic games for cooperation, Lee et al. [25] found that participants gave less to the AI when it could purportedly not reason than when the AI was described as being able to reason. Likewise, participants gave less to the AI when it could not feel than when the agent was described as being able to feel [25].

Another theme that emerged was the connection between mental state attribution and perceptions of morality and responsibility. For example, Hindennach et al. [19] examined the prevalence of mind-attributing language in explainable AI (XAI) methods, and scrutinized effects of such language use on awareness and responsibility perceptions. The authors found three different types of mind-attribution languages (i.e., metaphorical, awareness, & agency). Testing possible effects of these mind-attributing languages, Hindennach et al. [19] found that the language increased perceptions of awareness and responsibility. Importantly, language using mind-perception also concealed the responsibility of AI experts' involvement compared to non-mind-attributing explanations or a no explanation condition.

Varying levels of information about the AI's algorithm and the occurrence of a moral violation, Shank and DeSanti [39] found that people generally attributed responsibility to an AI, and perceived events of a moral violation as morally wrong. Based on this finding, the authors concluded that participants, hence, had to assume some levels of AI awareness, intentionality, and responsibility to judge it as morally wrong. In line with this, attributing mental states to the AI was positively associated with people's attribution of intentionality and wrongness.

Adding to these results on how users relate to agents as well as how they ascribe responsibility and morality, two studies related higher mental state attributions to both increased future use intentions [17] and increased continued interaction intentions [28]. Investigating the domain of visual art, Messingschlager and Appel [31] found that the attribution of mental states to generative AI predicted appreciation of the visual art piece.

Differential findings: In addition to the causes and consequences of attributing mental states, various works compared the potential for mental state attribution between different agents. In most cases, the comparison was drawn between an autonomous

²Some studies also reported correlational results for mental state attributions. However, as these were not the main focus of this work, further information about correlational results can be found in Appendix 6.3.

agent and a human. The overwhelming result was that people attributed less mental states to the autonomous agents than to humans. More precisely, participants reported to perceive autonomous agents to experience less pain and passion [2], perceived less warmth, competence, experience, and agency [11], ascribed less legal rights [2] and responsibility [19], and experienced less empathy towards them [29] as compared to other human beings.

Comparing different non-embodied autonomous agents, Scott et al. [38] found no difference in consciousness attribution for a vacuum robot, GPT-3, and Alexa. In addition, Srinivasan and Sarial-Abi [41] found that an anthropomorphic algorithm is punished more for an error than a non-anthropomorphic algorithm. Likewise, the authors could find that machine-learning algorithms were punished more than none-machine learning algorithms [41].

As a result of this reduced attribution of mental states, participants were, for example, more likely to adhere to advice given by a human [30] but harm caused by an algorithm was punished less than harm caused by a human [41].

Only one study reported equal ratings for humans and an autonomous agent: Messingschlager and Appel [31] could show that genAI art was equally appreciated as human art, although the AI artist was attributed less agency and less experience than a human artist.

4 Summary and Takeaways

In this systematic review, based on a literature body of 400+ papers, 26 empirical studies on users' mental state attribution to non-embodied, autonomous agents were reviewed. In a first step, it was analyzed which communities publish work on mental state attributions. In a second step, the study design choices (samples, tasks, agents, measures) were reviewed, and an overview of the studies' results provided. For each section, critical points are summarized and reflected below.

4.1 Current trends and challenges:

Research communities: The field of mental state attributions to non-embodied, autonomous agents is more actively explored within HCI-related fields such as psychology, communications science, information systems, and marketing. While this might seem understandable given the nature of the agents being predominantly chat- and voice-bots (see more below), it raises two questions: Why are not more people within the HCI community, and particularly within the biggest HCI conference, concerned with mental state attributions? And, what is the study of mental state attributions missing, if this community is not as involved?

Answering the first question, it is speculated that it might seem that findings from HCI-related fields such as psychology, communications science, information systems, and marketing are sufficient to inform HCI scholars. However, related to the second question, the relative absence of the HCI-community in the study of mental state attribution might be critical after all. As a highly interdisciplinary field, the HCI-scholars have the unique ability to address issues related to user interaction design, ethical considerations, and user experience, areas that are essential to understanding how people attribute mental states to agents—a potential that is now being largely overlooked.

Focus on chat- and voice bots: The reviewed research tends to focus on digital agents with specific functions—particularly chat- and voice-bots, which are primarily text-based agents. While these studies contribute valuable insights into mental state attribution in dialogue-based interactions, they do not capture the full diversity of autonomous agents that people encounter in digital environments. Notably absent from the literature are studies focusing on the attribution of mental states to agents where interaction is guided through more abstract interfaces such as content recommendation and content moderation, or image recognition. These types of agents may present unique challenges for mental state attribution due to their lack of a conversational interface and their more implicit interactions with users. Future research should consider incorporating a wider range of agent types and consider how the design and functionality of these agents influence mental state attribution.

Assessing mental state attribution - questions of validity: One of the most concerning findings from this review is the lack of consistency and rigor in the measurement of mental state attributions. Many studies did not use validated scales or established measures, which significantly hinders the interpretability and replicability of findings. The absence of reliable tools to measure mental state attribution limits the ability to draw firm conclusions about the phenomenon and makes it difficult to compare results across studies. This highlights the need for the development and adoption of standardized, validated scales to assess how users attribute mental states to non-embodied agents. Without such tools, the research community risks producing findings that are not only difficult to interpret but also unlikely to be replicated or extended.

Central findings regarding mental state attribution: The reviewed studies consistently show that when users attribute mental states to autonomous agents, they tend to lower their guard and become more vulnerable to influence. This is particularly concerning in contexts such as marketing or persuasion, where agents may subtly shape user decisions or opinions. Moreover, users may shift their perception of responsibility, attributing more accountability to the agent rather than recognizing responsibility of other parties such as designers and developers. This shift in responsibility could have serious implications, especially as non-embodied agents become more autonomous and integrated into decision-making processes across various domains.

The consequences of mental state attribution suggest a need for greater ethical consideration in the design of non-embodied agents. Designers should be aware of how their agents are perceived by users and ensure that users maintain an appropriate sense of agency and responsibility in interactions. Future research should investigate the long-term effects of mental state attribution on users' cognitive and behavioral patterns, and explore how these effects may vary across different types of agents and user contexts.

4.2 Limitations

This systematic review has a few limitations regarding the overall scope. First, preprints and qualitative studies were not included in the search strategy. Particularly excluding preprints did not allow to capture the most recent research, as preprints often represent the latest findings in the field. Second, the search was very focused

by including only works that directly discussed "mental states" and/or "mind attribution". While focused, this may have excluded relevant studies that examined more specific aspects of mental state attribution, such as the attribution of emotions, intelligence, or anthropomorphism. This narrower scope may have led to the omission of studies that address related but distinct facets of mental state attribution.

5 Conclusions and Future Directions

This systematic review provides a deep-dive into the current state of research on mental state attribution to non-embodied autonomous agents. While significant insights have been gained, there are still many unanswered questions and gaps in the literature. Future research should aim to broaden the scope of agents studied, include more diverse methodological approaches, and prioritize the development of validated measurement tools. Additionally, there is a pressing need to investigate the ethical implications of mental state attribution, particularly regarding user vulnerability and shifts in responsibility. By addressing these challenges, future research can provide a more comprehensive understanding of how users interact with non-embodied agents and how these interactions shape their perceptions and behaviors.

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6 Appendix

6.1 A

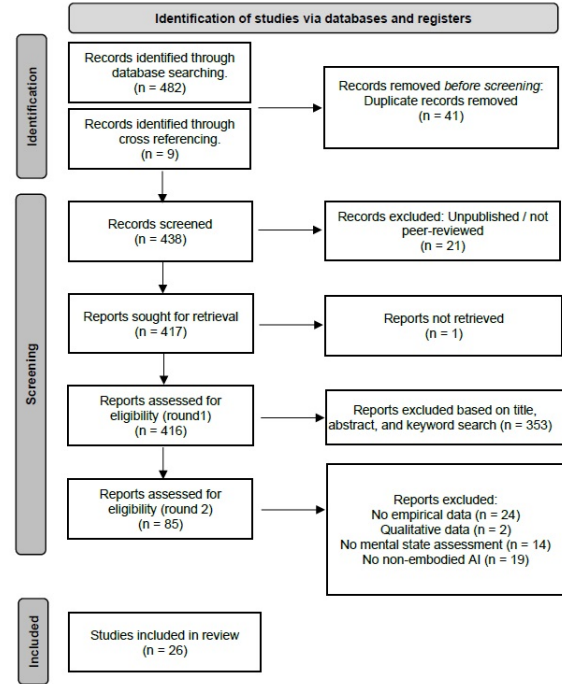


Figure 1: Flow chart of the paper selection process.

6.2 B: Additional measures complementing mental state attributions

In addition to the assessment of mental state attributions, many studies included other measures such as the perceived humanness or anthropomorphism of the agent [4, 5, 41], perceptions of predictability [4], intelligence [5, 7, 29], animacy [5], autonomy of the system [11, 21], wrongness, and responsibility of the respective agent [39, 41]. Following theoretical considerations of the Technology Acceptance Model and the Unified Theory of Acceptance and Use of Technology, one study assessed the constructs' perceived usefulness, perceived ease of use, enjoyment, and social presence [34]

Additionally, some studies included a measure of future intentions to use the agent ([21, 26, 28, 34]) as well as evaluative measures like trust [34] or appreciation of an art piece [31]. Interindividual measures encompassed users tendency to anthropomorphise [19], social dominance and authority preferences [17], social anxiety, loneliness, rumination, problematic use of conversational AI [20], attitudes towards AI [7], technical affinity [11], familiarity with the agent [7, 38], and frequency of use [22].

Besides the studies that employed economic games [4, 25], only one other study recorded user behavior (here: call duration with a voice-bot, see [29])

6.3 C: Correlational findings

Unlike the results reported in the previous two paragraphs on causes and consequences of mind perceptions, the studies presented here followed a correlational set-up. In all cases, different variables were measured and their association with each other was analyzed. However, because none of the following works included experimental variations, results must be interpreted in a correlational manner, despite some works portraying them as causal.

The findings across the reviewed studies suggest that previous experience with an autonomous non-embodied agent plays a significant role in shaping perceptions and behavioral intentions. For instance, perceived anthropomorphism and perceived intelligence were positively associated with higher purchasing intentions, but this effect was observed only in individuals with greater prior experience using the agent [5]. Similarly, having prior experience specifically with ChatGPT was found to increase perceptions of the chatbot as being conscious, demonstrating the influence of familiarity on mental state attribution [7].

Moreover, mental state attribution was found to be associated with various related constructs, including perceived animacy, mind perception, and artificial autonomy. While perceived animacy increased users' positive attitudes toward the agent, it did not translate into higher purchasing intentions [5]. Mind perception emerged as a significant moderator in user experiences: it amplified the positive relationship between social anxiety and problematic use of conversational AI (PUCAI) and mitigated the impact of rumination on PUCAI [20].

The perception of artificial autonomy was shown to enhance evaluations of both competence and warmth in intelligent personal assistants (IPAs). These evaluations, in turn, positively influenced users' intentions to continue using the agents. However, competence and warmth had differing effects: the influence of competence was stronger than warmth. Further analyses revealed that sensing and action autonomy played a more significant role in shaping perceptions of competence than warmth [21].

Interestingly, other dimensions of mental state attribution yielded mixed findings. For instance, perspective-taking ability was not associated with Theory of Mind (ToM) abilities, whereas higher predictability in agent behavior was linked to lower levels of anthropomorphism [4].

Mental state attribution also demonstrated associations with constructs that might initially seem unrelated, such as trust, attitude, and social health benefits. Social cognition, specifically the competence dimension, was found to have a strong positive effect on trust and attitudes toward AI agents. Inferred social cognition, particularly with respect to perceived competence, emerged as a key factor in fostering trust toward AI systems [34].

Perceptions of human likeness and mind in chatbots were linked to social health benefits. Individuals who viewed chatbots as more humanlike and attributed higher levels of consciousness, experience, and agency to them also perceived greater social health benefits from engaging with these systems [16].

One reviewed study [7] also provides insights into how people attribute mental states to ChatGPT, specifically. While approximately one-third of participants stated that ChatGPT could never possess consciousness, the majority believed that some level of

consciousness might be possible. Among these, those who thought consciousness was possible tended to conceptualize it more in terms of experience than intelligence. Moreover, participants who attributed higher levels of phenomenal consciousness to ChatGPT also attributed more experience-related mental states than intelligence-related ones [7].

Interestingly, participants systematically overestimated the extent to which others would view ChatGPT as conscious. Additionally, ChatGPT was predominantly seen as more capable of intelligence than experience, highlighting a distinction between cognitive and experiential mental state attributions in this context [7].